torch.logdet#

https://pytorch.org/docs/stable/generated/torch.logdet.html#torch.logdet

Description:

Calculates log determinant of a square matrix or batches of square matrices. It returns -inf if the input has a determinant of zero, and NaN if it has a negative determinant. Backward through logdet() internally uses SVD results when input is not invertible. In this case, double backward through logdet() will be unstable in when input doesn't have distinct singular values. See torch.linalg.svd() for details.

Function Signature

```
torch.logdet(input) → Tensor#

Parameters
```

Code Examples

```
>>> A = torch.randn(3, 3)
>>> torch.det(A)
tensor(0.2611)
>>> torch.logdet(A)
tensor(-1.3430)
>>> A
tensor([[[ 0.9254, -0.6213],
         [-0.5787,  1.6843]],

        [[ 0.3242, -0.9665],
         [ 0.4539, -0.0887]],

        [[ 1.1336, -0.4025],
         [-0.7089,  0.9032]]])
>>> A.det()
tensor([1.1990, 0.4099, 0.7386])
>>> A.det().log()
tensor([ 0.1815, -0.8917, -0.3031])
```

Notes & Warnings

NOTE

Note Backward through logdet() internally uses SVD results when input is not invertible. In this case, double backward through logdet() will be unstable in when input doesn't have distinct singular values. See torch.linalg.svd() for details.

NOTE

See also torch.linalg.slogdet() computes the sign (resp. angle) and natural logarithm of the absolute value of the determinant of real-valued (resp. complex) square matrices.

Detailed Documentation

Documentation

Calculates log determinant of a square matrix or batches of square matrices.

It returns -inf if the input has a determinant of zero, and NaN if it has a negative determinant.

Backward through logdet() internally uses SVD results when input is not invertible. In this case, double backward through logdet() will be unstable in when input doesn't have distinct singular values. See torch.linalg.svd() for details.

torch.linalg.slogdet() computes the sign (resp. angle) and natural logarithm of the absolute value of the determinant of real-valued (resp. complex) square matrices.

input (Tensor) – the input tensor of size (*, n, n) where * is zero or more batch dimensions.